

## Configuration-level split aliasing in object-detection benchmarks: what the reported split is selected on, and how much of the selection survives an independent split

### Abstract

Object-detection papers report a number from a split they call held out. We audit that premise, in three results.

**1. The convention is broken in our own corpora.** 6 of 14 corpora give the reported number no independent held-out split: the checkpoint was selected there. Training and evaluation do not overlap — **checkpoint selection**, not contamination.

**2. The direction varies by corpus.** The premium is **positive in all four cells** (§5.1; the baseline arm gains more), the **paired  $\Delta$ gap** changing sign by corpus (**not the selection term**); a declared, **descriptive** 13-cell family leaves **3** after BH; **five current-generation coders** agree at  $\kappa$  **0.824** and, re-verifying each row, **independently reach 19 of 30 corrected cells** (eleven unanimously), **26 of 76** stay undecidable; the printed **12–14 of 19** rest on **one rater**, **10–11** after re-verification (§6.1, S12). **The mean direction is metric-robust; one cell's significance and sign unanimity are not** (smoke2sf +0.063 pp,  $p = 0.54$ , 6+/5–).

**3. Partial realization on a disjoint split.** Separating the selection set from the reported set, the premium is realized only in part: the **claim-level interval includes zero** ([–17.3 %, 53.2 %],  $P = 0.118$ ); on a difference scale **–1.367 pp** (CI [–1.59, –1.15]); run-level **[8.4 %, 37.4 %]** ( $P = 0.002$ , a bootstrap tail proportion) and leave-one-out **9.77 %–34.16 %**; the arm-cell and run-weighted point estimates are **descriptive**, given only with intervals (Table 4).

**Replication lowers it 9 %–29 %** (§8); no power analysis was registered, so the nine  $n = 3$  cells stay descriptive.

**Keywords:** evaluation validity; dataset splits; checkpoint selection; object detection; benchmark audit; reproducibility

### 1. Introduction

A detection paper reports mAP50-95 on a split it calls held out, implying a number that estimates performance on unseen data. The inference needs a premise the paper rarely states: the checkpoint was not chosen on that same split. Binding `val:` and `test:` to one directory breaks it silently — the pipeline

evaluates there, keeps the best-scoring checkpoint and reports its score: a **maximum**, not a sample. Surveys of the detection literature [1] trace how that number became the field's unit of comparison; this paper audits the reporting convention producing it.

That premise is audited in three places, in increasing distance from our own work: **our own corpora**, where we count how many of our corpora alias the two keys (19 YAML files, counting by file); **19 public benchmarks**, audited row by row for which split the reported number comes from; and **20 published third-party training logs**, where we measure the premium directly, in data we did not produce.

Three further things we have not seen done here. We **measure the premium's fate** under a protocol separating the selection set from the reported set (**best** and **last** checkpoints and both splits exist, so the full  $2 \times 2$  grid is computable): how much of the **val**-selected premium is realized on an independent split — the measurement this paper contributes and §3.1 formalises. The protocol is the companion's and is **cited, not restated**. And we **pre-register** a generalization hypothesis, then report that it misses its criterion — a registration **also reported in the companion**.

### 1.1 Why the unit matters

Ask "how many of these benchmarks have a split problem?" and, unless a unit is declared, every answer is defensible and they disagree by a factor of six: 2/19 (**structural**: one path bound to both keys), 10/19 (**'release' unit**: no independent held-out test), 13/19 (**'protocol' unit**: local independent testing gated), **12–14 of the 19** (**'reported' unit** — **12** the single rater's printed coding, **14** that rater's rows after a **non-blind recoding** moved two; **five blind coders agree at  $\kappa$  0.824** and, once the record is re-verified first-hand, independently reach **19 of the 30** corrected cells, eleven unanimously (**S12**); the reported number's split is non-independent). The readings above are all defensible, and §6.1's four counting units (**release**, **protocol**, **yolo\_dist**, **reported**) are the **columns** they come from — including **yolo\_dist**, omitted above, at **4/19**. Declaring the unit is therefore a methodological premise, not a caveat (§3, §6).

### 1.2 What we do not claim

We do not claim that aliasing is data contamination: the image-level intersection between the training and held-out evaluation images is **zero** (SHWD excepted,

where `train` and `val` are one directory). We do not claim a single bias direction: our paired analysis finds **two corpora where `val` understates and two where it overstates**, and **nine cells are not testable at  $n = 3$**  (§5.1, §7.1). Nor do we claim the convention is framework-general (§6.5) or that the audited benchmarks are wrong: they report what their configuration computes.

**Positioning (one sentence).** We claim neither to be first to audit split problems in detection benchmarks nor to audit image-level leakage: both are taken. We occupy a different cell — **configuration-level aliasing, a direction that varies by corpus, and the requirement that a count declare its unit**. Every number points to a local artifact; every claim narrowed after external checking is recorded as narrowed (§5.4).

**Scope.** All quantitative claims about selection behaviour come from **YOLO-lineage pipelines**, the only family among the six frameworks we checked (five pinned repositories plus the Ultralytics release) selecting the checkpoint by the validation metric by default (§6.5) — stated at the claim, not only in a limitations section. **The registered multi-target replication is reported in the companion paper, and its first report, criterion and verdict stay there** (§9): this paper cites it for readability and **claims none of its results**.

## 2. Contributions and related work

**Where this paper sits.** Reporting practice has itself been audited: leakage and near-duplicate contamination [3], [4], [11]; split naming and the choice of standard splits [8], [9]; experiment- reporting standards [7]; checkpoint-selection criteria [6]; and budget-dependent rankings [2]. Each audits **data, reporting, or selection; none audits a release’s configuration binding** — the object of §3–§6.

### 2.1 The closest work, and where this paper sits (detailed)

- > The detailed comparison — the nine prior lines of work, what each measures, and the axis on which
- > this paper differs — is **Table S45 in the Supplementary material**; it is not printed here because
- > the journal counts text inside the 20–35 page limit and the comparison is background, not a claim.
- > **§2.2 (Contributions) and §2.3 (what we do not claim) are unaffected.**

## 2.2 Contributions

**What this paper reports first** (the boundary to the companion is in §2.3): the configuration-level audit of the `val/test` convention and its selection premium (§4–5); and the **measured** realization of that premium on a disjoint split with the reporting conventions it depends on (§8.6–8.7, §11, §3.1).

1. **We quantify the magnitude and direction of configuration-level aliasing.** The selection premium (best – last) is **positive in all four cells, the baseline arm gaining more (near-definitional; §5.1; four paired 95 % CIs excluding zero, §5.1 Tier 1) — consistent with the selection-bias literature [5], not new in kind. All four cells meet the stated-n criterion (n = 10–11)** and all four survive a Benjamini–Hochberg step over **these four** at  $q = 0.05$  — the fourth, `dota15`, entered at  $n = 3$  and was escalated to  $n = 10$  under a criterion fixed in advance (**Table S28's note**) — so the convention systematically shifts **arm-to-arm comparisons**; its direction varies by corpus: three-way paired differencing gives 2 cells where `val` understates, 2 where it overstates and 9 not testable at  $n = 3$  — the **no-systematic-direction** [10] reports, measured here on detection corpora.

**We do not claim a general understatement** (§5). It appears in third-party logs too (a searched 20-log sample; §5.5) — corroboration that the component exists elsewhere, **not independent evidence** and not a field-level rate.

*Note: image-level\** leakage magnitude is quantified by the two 2026 studies; **configuration-level** aliasing magnitude has not been measured before.\*

2. **We measure how far the selection premium transfers.** The measurement is the contribution; the protocol that makes it readable is the companion's and is **cited, not restated** here (§8.1). On the five arm-cells measured here the `val`-selected premium is realized on the disjoint split at

**23.3 %** with all five arm-cells at  $n = 10$ , and it is **not a constant**: the rate falls with the validation noise of the criterion used to select (§8.6–§8.7). The identity tying it to the winner's-curse share we **adopt** [13], not claim; the closest concurrent work [12, 15] reaches the same **direction** with a different quantity, and neither reports a **ratio of difference premiums** —  $(T_{\text{best}} - T_{\text{final}})/(V_{\text{max}} - V_{\text{final}})$ , on disjoint splits, in detection, on our own arm-cells, with measured `val` noise as moderator. A same-estimand ratio does exist elsewhere — in genomics, where it is written as a relative bias and

measured at 10–60 % (**Supplementary S12** carries the full citation). The replicate axis is **data order** (§8.4) — the **seed-budget discipline reproduced inside this setting** rather than a new one.

3. **We disclose our own defects, and the corrections that followed** rather than removing them: our analysis' first version had non-reproducible items — pod-dependent scripts, a citation to a file not holding the quoted numbers, no written endpoint convention. §10, not deleted.

### 2.3 What we explicitly do not claim (shared tooling)

The companion and this paper share project-internal process protocols — a multi-model review checklist and a local/cloud scripting discipline — and neither claims originality for them: **Supplementary S6, Shared process protocols**.

### 3. Units: the methodological premise

"Have a split problem" is a property of a (**benchmark, counting unit**) pair, not of a benchmark. [14] **varies the definition on one file; we vary who controls the split and measure a different quantity: the share of `val`-selected premium that survives a disjoint split**. We use four units, defined by **who controls the split**:

- **`release`** — the split the benchmark's **own published release** ships: the `train:` / `val:` / `test:` entries and paths in the distributed archive. *Example:* an archive whose `data.yaml` binds `val:` and `test:` to the **same directory** records "yes" here (§4).

- **`protocol`** — the split the benchmark's **official protocol** tells you to use, which need not be the released one: a protocol may carve its own validation set out of the released training split.

*Example:* an aliasing release whose documented protocol still requires a held-out test split records "no" here and "yes" above — the units answer different questions.

- **`yolo\_dist`** — the split a **generic YOLO distribution package's** own configuration resolves to, i.e. what a practitioner taking the pipeline off the shelf gets. *Example:* a clean release whose community YAML points `val:` and `test:` at one directory records "yes" **only** here.

- **`reported`** — the split the number **the literature prints** came from (not the official leaderboard's own score), read from the release documentation and the

dataset's own paper. *Example*: two papers on one benchmark can differ here while agreeing under all three units above — hence the headline count uses this unit (§6.1).

**A count is meaningless without its unit**: the same nineteen benchmarks answer differently under each (Fig. 1).

**The proposition is not ours to claim**. That *the same data yield different defensible audit counts under different operational definitions* is stated independently for molecular benchmark auditing, in a 2026 preprint whose abstract closes "different defensible definitions return different answers from the same file" [14]. **The observation is the cited literature's; its instantiation in detection evaluation, and the four named units, are ours** — the units named by *who controls the split*, the 19-row evidence table, and the premium on **five arm-cells**. It is orthogonal to [15], which separates leakage by **type**; we separate the **counting subject**, so one type can still be counted under four units (§6.1).

> **Table S2** → Supplementary §S8.

**The 'reported' unit is read from the release documentation and from the dataset's own paper**: we did not sample papers per benchmark, so no prevalence claim is made (§11). For each of 19 benchmarks we record all four units, an evidence class and a confidence level, and recompute every percentage from the row markers under a per-unit subtotal check **and one stated 'n\_a' rule (Supplementary S12)** — the full  $19 \times 4$  table and the cross-check against the original audit are **Supplementary S1**, the derived counts §6.1 and its table.

**The rule itself is not new**. The same toolchain's own course states it normatively — "*Test set is touched once, at the end. Not for tuning.*" and "*The splits must be independent on the dimension that matters at deployment.*" [16] — so what this paper adds is not the rule but the **audit and the count** under it.

**The premise we rely on throughout**: a "yes" under any unit means *the reported number is not a sample from an independent split*. **What that does and does not license is stated where the counts are printed** (§6.1 and the note under its table).

### 3.1 Formalisation: what the four quantities are, and what they can and cannot identify

**Setup**. Training is a randomised map from (seed  $\omega$ , recipe  $\theta$ , checkpoint  $e$ ) to weights, and evaluation a fixed functional on a labelled split; write  $V(e)$  and  $T(e)$

for a checkpoint's `val` and `test` readings and  $\hat{e}$  for the `val` argmax. Sections 5–9 measure four quantities: the selection premium,  $\Delta_{\text{gap}}$ ,  $\kappa$ , the realization rate (**notation and formal statements: Supplementary S11**).

**Definition — the raw gap is not identifiable from the pair alone.**  $G = V(\hat{e}) - T(\hat{e}) = \delta + o$  mixes a split-difficulty term with a selection-optimism term; the decomposition, **and the reading that a single-model endpoint pair is unable to separate its two terms, are [17]'s**, while its **strong form** — that *any*  $\delta'$  fits the observation equally well — is **stated here as a formalisation rather than attributed to that literature. The pairing assumption is what makes  $\Delta_{\text{gap}}$  meaningful**, failing by construction under aliasing so that §9 cannot be read as a clean-protocol result; the identity holds value by value on all 13 scanned cells and both escalated cells at  $n = 10$  (§5.6).

**Adopted identity, not a claim of this paper.** With  $V(e) = v(e) + \varepsilon_e$  and  $T(e) = v(e) + \tau + \eta_e$ ,  $E[\text{premium\_test}] = E[\text{premium\_val}] - WC$ ,  $WC := E[\varepsilon_{\hat{e}} - \varepsilon_{e_f}] \geq 0$  — the argmax is over a **candidate set fixed in advance** and  $e_f$  is a **pre-specified member of it** (what the budget-fixed `last` gives) — so the realization rate is  $1 - WC / E[\text{premium\_val}] \leq 1$ , with equality only if selection is driven by signal alone [13] (**Supplementary S11** states which part of that the archive can and cannot settle) — the optimizer's curse in this paper's notation, the frame for §8.6–§8.7. **Two consequences are testable on the archive** — the per-run inequality, and monotonicity in the `val` noise (**Supplementary S11**). A third form, that the premium scales as  $\sigma\sqrt{2 \ln E}$ , was **tested and withdrawn** (**Supplementary S11**); contemporaneous work measures  $\approx 5\times$  on the same  $\sigma\sqrt{2 \ln K}$  bound [18].

**Corollary (the premium is mostly shape, and the peak is the worst place to stand).**  $\text{premium\_r} = \text{shape}(E) + v_r$  holds **exactly**,  $\text{shape}(E)$  being the mean curve's peak-to-final gap, so  $\text{shape}$  is a functional of the mean curve alone — **not** an independent check on  $\sigma$  — and  $E[v_r] \geq 0$ . On our 40-run four-budget series **the monotone criterion is met on the clean re-run** while the  $\sigma\sqrt{2 \ln E}$  form is **outside** its pre-declared band, so that scaling **stays withdrawn**; the pre-declared shape control passes on the mean it specifies but fails per arm, and we report both readings — the with-early-stopping series carrying the pre-declared verdict — so **we do not claim an 'E' effect**. The checkpoint rule is a **reporting rule** (the top-5 mean beats the argmax on the

same 40 runs), with **three limits**: an empirical capture rate, a `val`-side cost the `argmax` does not carry, and a two-split, five-epoch-grid scope.

**What is ours, and what is not.** Of the statements that follow, the raw gap's definition [17], the impossibility result for endpoint-pair estimators [19] and the family-dependence procedure [20] are **machinery rather than results of ours**, and the identity is **adopted** from the selection-bias literature; **the corollary alone carries measurements of this paper**. Full statements and proofs are in **Supplementary S11**, and the detail displaced from here in **Supplementary S12**.

### 3.2 Six experiments, five judged criteria, on GPU machines, each with its criterion written down first

The propositions of §S11 are testable, so we ran them on rented GPUs with the **criterion fixed before the runs**; **Supplementary Table S37** gives, for each of the six experiments, the criterion it was judged against. Counts below are over **experiments**, not criteria.

**The two negative outcomes and the redone design are stated as such in Supplementary Table S37** (G1 is void; **two of the five judged experiments are negative** — G1' and G2; the values behind every verdict are in **S31–S33**, the G1 family in **S12**; the reading below is stated more weakly in **Supplementary S11, section 13.3**).

**We then ran the clean test** of G3's epoch-budget series on **one machine with early stopping disabled**, all four budgets present: C1 is **met**, C2 falls **outside** its band, and C3 **passes** on the mean  $\hat{e}/E$  it specifies but not per arm — see §3.1 and **Table S34b**; **the pre-declared verdict stays with the with-early-stopping series**. The judged criteria carry one message, **at the level of sign, not of functional form**: the selection bias moves with the amount of selection noise and with the number of selection opportunities, but not with whether the two splits are the same file — **the form itself was withdrawn** (§3.1). Two consequences do not follow: §4's configuration-level fact stands, while the stronger reading — that aliasing *adds* a measurable amount on top — does not. **The premium this paper measures is the premium of selecting on the reported split; aliasing is the configuration that guarantees it, not an additional term.**

On the shared 5-epoch grid the `val` and `test` peaks agree in only **17.5 %** of runs, and conditioning on that single observable splits the 40 runs: where they

agree the **median  $\Delta$ gap is lower** (mean **+0.13 pp**, median **-0.20 pp**,  $n = 7$ ; **post-hoc**), where they differ it is **systematically not** (mean **+0.91 pp**,  $n = 33$ ; permutation  $p = 0.009$ , **post-hoc**; the median reading is  $p = 0.0011$ ). **This is a partial predictor, not an explanation of the sign**; its limits — including the four of seven agreement runs whose gap is non-positive — and the per-run table and test are in **Supplementary S11**.

#### 4. Finding 1: our own corpora

##### 4.1 How common the aliasing is

In **our own packaged configurations** — the YAMLs we train from, a different object from a benchmark's own release — **6 of 14** give the reported number no independent held-out split: **AI-TOD, DOTA v1, DOTA v1.5, Mendeley face-mask, SFCHD** (five binding `val:` to `test:`) and **SHWD** (which binds `train` to `val`; **S1** row 13). Of the remaining 8, **4** keep the keys genuinely distinct and **4** have no `test:` key at all — the three groups account for all 14, and this is a **count over the configurations we hold, not a prevalence, nor the benchmarks' own releases** (audited separately, §6.1). Counting by file, **19 YAML files** alias the two keys, in five locations; on the archived manifest subset the ratio is **8 of 28**. **These three bases answer three different questions** — configurations (6 of 14), the files inside them (19), the archived manifest's rows (8 of 28) — **not three estimates of one rate**.

**For the corpora whose `val` and `test` are the same directory this is not contamination**: their image-level intersection with training is **exactly zero**, so the aliasing changes not *what* was evaluated but *which checkpoint* was reported, turning "selection contamination" from an ethical question into a **measurement** question. **SHWD is the exception and is not folded into that statement**: with `train` and `val` on one directory, its training and evaluation images are the same files. **It still counts among the six, for the reason the list is about: its `val` is not an independent held-out split either**.

##### 4.2 The failure is produced by a tool, not by carelessness

A PASCAL VOC [21]→YOLO conversion script wrote two keys from one `images_dir`, so `train == val` appears **because the generator could not produce distinct values** (§6.1 records the same for Mendeley face-mask [22]). §10's first recommendation: make "are `val` and `test` the same source?" a **pipeline-level hard check** — human YAML review will not catch it.

## 5. Finding 2: the selection premium — magnitude, direction, and mechanism

**Evidence.** `eval_validity/selection_premium_local_20260916.txt`, reading **569 archived runs** from the local archive (the local version is the 2026-09-16 repair).

**Definition.**  $\text{premium} = \text{best-epoch} - \text{final-epoch}$  mAP50-95 **within a single run**. When `val == test`, "best" was selected on the reported split, while `last` and `last5` are endpoints **no selection rule can reach**. This is the checkpoint-selection term of the run-to-run variation studied in [23] (early stopping, data order) and [24] (benchmark variance across seeds). Being a maximum over a noisy curve, it also carries the upward bias of any selected extreme [13]: what those works average over as uncertainty, we measure as a reproducible component. **The two arms.** Every cell is a paired comparison in which the **baseline** arm follows the published convention and the **strategy** arm is the intervention under test — in the cells reported here, raising the peak learning rate from **0.001 to 0.005** with everything else held fixed (batch 32, `imgsz 640`, SGD; YOLOv12n with the shape-aware IoU). The two arms run the same code and the same data.

> **Table S3** → Supplementary §S8.

### 5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus

The direction claim **must be reported in two tiers**: "the convention understates the effect" is wrong, and external checking refuted it (§5.4, II). **Endpoint convention (frozen):** `best - last`

**within a single run**, both endpoints reported for every cell; §7.2–§7.3 quantify the other convention.

**Tier 1: the selection premium is positive in all four cells and the baseline arm gains more**, consistent with the selection-bias literature, not new in kind; **on the monitored split that positivity is near-definitional** (best is that curve's own argmax), and **the finding we report is the paired arm difference and its test side (§8.6)**. Pairing brings **no** variance benefit: the paired *t* of 5.74 and the unpaired Welch *t* [25] of 5.68 agree — at  $p = 0$ , **standard arithmetic rather than an empirical finding of ours** [24]. **Permutation checks** agree with the *t*-tests at each cell's attainable floor (**Supplementary S12**).

**Meta-analyses of published results have long quantified this optimism:** development-set early stopping is measured to *overestimate realistic model performance*, by as much as **18.0 %** accuracy [26]; and a large Kaggle meta-analysis finds *little evidence of substantial overfitting* [27] — complementary rather than contrary, since its mechanism is holdout **reuse** and ours needs none.

**Metric sensitivity.** On `metrics/mAP50(B)` the mean direction holds in all four cells, but `smoke2sf` loses both its significance (**+0.063 pp**,  $p = 0.54$ ) and its sign unanimity (6+/5−, not 11+/0−): **the mean direction is metric-robust, one cell's significance and sign unanimity are not** (full mAP50 columns and both mAP75 columns — the mAP75 val side is a **5-epoch grid**, so a rate computed from it is **upward-biased** — **Supplementary S12**).

**Table 1.** Tier 1 — each arm's **peak–final gap** and the **paired difference** between arms, by cell; the gap is a within-run maximum (§5.1), so the finding is the paired difference. `n` counts paired units (ten file-order replicates, one unseeded — **Supplementary S12**); both differences hold at  $n = 10$ . **A different run set from the transfer measurement in §8.6:** these are the 569-run local archive under the published (aliased) protocol, while that measurement's premiums come from the clean three-way runs — the two tables' premiums are different quantities, not two readings of one. Source: `verify_two_critiques_20260916.txt`; the `dota15` row's  $n = 10$  escalation is in Table S28.

| Cell                            | n<br>(paired<br>units) | peak–<br>final<br>gap,<br>base | peak–<br>final<br>gap,<br>strat | **paired<br>diff** | sd        | t        | p                        | per<br>m.<br>*p*<br>(floor<br>2/2 <sup>n</sup> ) | 95 %<br>CI                  |
|---------------------------------|------------------------|--------------------------------|---------------------------------|--------------------|-----------|----------|--------------------------|--------------------------------------------------|-----------------------------|
| smoke2sf (strong)               | 11                     | +0.79<br>9                     | +0.38<br>2                      | **+0.4<br>16**     | 0.2<br>40 | 5.<br>74 | 1.9×<br>10 <sup>−4</sup> | 0.00<br>098                                      | [+0.<br>255,<br>+0.5<br>78] |
| shwd2sf<br>(corroborating)      | 11                     | +1.03<br>8                     | +0.51<br>2                      | **+0.5<br>26**     | 0.2<br>07 | 8.<br>41 | 7.6×<br>10 <sup>−6</sup> | 0.00<br>098                                      | [+0.<br>386,<br>+0.6<br>65] |
| a2d15<br>(directional<br>probe) | 10                     | +0.53<br>4                     | +0.26<br>2                      | **+0.2<br>72**     | 0.1<br>83 | 4.<br>69 | 1.1×<br>10 <sup>−3</sup> | 0.00<br>391                                      | [+0.<br>141,<br>+0.4<br>03] |
| dota15                          | 10                     | +0.94                          | +0.49                           | **+0.4             | 0.1       | 7.       | 4.9×                     | 0.00                                             | [+0.                        |

(within-domain) 1 2 49\*\* 96 23 10<sup>-5</sup> 195 309, +0.589]

**Table 2.** Tier 2 — the paired difference  $\Delta_{\text{gap}}$  by corpus.  $\Delta_{\text{gap}} > 0$  means `val` understates the strategy arm’s effect.  **$\Delta_{\text{gap}}$  is the split-gap arm difference**, not the selection effect: it decomposes into a difficulty gap and a selection term, and  **$\Delta_{\text{gap}}$ ’s sign is not the selection term’s sign (Fig. S1)**. Equal split difficulty across arms is an **assumption we state**; the selection contribution is the **selection-action contrast**  $\Delta_{\text{prem\_val}} - \Delta_{\text{prem\_test}}$ . ‘**Reading**’ names  **$\Delta_{\text{gap}}$ ’s sign, not the selection term’s** (smoke2sf:  $\Delta_{\text{gap}}$  +0.461, selection −0.580). Sources: `xeval_analysis_20260916.txt` (the two `val`-understating cells) and `x1_tier* / teval.csv` (the two `val`-overstating cells).

| Corpus     | n   | $\Delta_{\text{gap}}$ (pp) | p                    | 95 % CI          | Reading (** $\Delta_{\text{gap}}$ ’s sign**) |
|------------|-----|----------------------------|----------------------|------------------|----------------------------------------------|
| ---        | --- | ---                        | ---                  | ---              | ---                                          |
| shwd2sf    | 10  | **+0.779**                 | 0.0053               | [+0.297, +1.261] | `val`<br>**understates**                     |
| smoke2sf   | 10  | **+0.461**                 | 0.0263               | [+0.068, +0.855] | `val`<br>**understates**                     |
| p_aitovis  | 10  | **−1.226**                 | 1.3×10 <sup>−8</sup> | [−1.370, −1.081] | `val`<br>**overstates**                      |
| p_vistod15 | 10  | **−3.315**                 | 6.5×10 <sup>−9</sup> | [−3.676, −2.953] | `val`<br>**overstates**                      |

> **The split of  $\Delta_{\text{gap}}$  is recorded for two of the four cells:** difficulty +0.759 / selection +0.020 (shwd2sf), and difficulty +1.041 / selection −0.580 (smoke2sf) — in the second the selection term is **opposite in sign to  $\Delta_{\text{gap}}$** , so the sign does not carry it, and **both intervals include zero (Supplementary S9, Fig. S1)**. The two `p_` cells’ `prem_test` was **not recorded at that time**: their per-epoch test curves were evaluated on 2026-09-30, so nothing is imputed and the values are now recorded (Supplementary S12, Table S38).

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> **The four rows below are an escalated base, not a sample:** the two `p_` cells reached `n` = 10 because their `n` = 3  $\Delta_{\text{gap}}$  had been flagged, so the 2/2 balance is a property of the escalation rule; the other nine cells are **descriptive (Supplementary S12)**. This cell is also the one where **3 runs** changed training path silently under GPU memory pressure, and why we do not call it clean is in §10, defect 10.

(Third §5 table, S5 → Supplementary §S8.)

**Tier 2: on the same three-way corpora the paired difference  $\Delta_{\text{gap}}$  varies in sign.** The raw  $M_{\text{val}}(\hat{e}) - M_{\text{test}}(\hat{e})$  is **not identifiable**; the paired difference removes the difficulty term **under the equal-difficulty assumption** (Table 2) —

**$\Delta_{\text{gap}} > 0 \Rightarrow$  ‘val’ understates. Its effective base is four cells, not thirteen,** and an **escalated base** rather than a random sample, for the reason Table 2’s note gives; **“2 understate / 2 overstate” is therefore a statement about these four cells, not a rate for the family.** No prospective power analysis was registered; the scan is in **Supplementary S12**.

**Both escalations met the criteria fixed in advance.** dota15 +0.449 pp (10+/0–); p\_aitovis –1.226 pp and p\_vistod15 –3.315 pp (0+/10–), permutation p = **0.001953**, the attainable floor at n = 10. The best.pt endpoint was identified, not assumed: on test it reproduces the archived sio\_b\_results.csv to **0.0046 pp**.

**The formal claim:** the convention systematically shifts **arm-to-arm comparisons** (Tier 1: four cells, CIs excluding zero); **the direction of that shift varies by corpus** (Tier 2: 2 understate / 2 overstate on the four-cell base) — **neither “a general understatement” nor “no systematic direction”**. BH at q = 0.05 **now leaves three cells significant** — p\_vistod15, p\_aitovis, shwd2sf [28], **reversing what we reported before the escalation.** **The two families are not the same test:** the **13-cell scan is descriptive**, the **confirmatory family is the four escalated cells** (declared before their runs, §3.2) — the direction statement rests on those four.

**Scope (cross-framework boundary).** The convention this paper audits belongs to the **YOLO lineage** — Ultralytics YOLO [16] and YOLOX [29]: at pinned commits, Detectron2 (TEST.EVAL\_PERIOD = 0) [30], MMDetection [31], DETR [32] and PaddleDetection [33] **report the final epoch instead** (§6.5) — that pipeline family’s convention, **not the field’s**.

## 5.2 Two versions must be disclosed side by side

**Two archives, two versions of this table.** The main text uses the **local archive** (569 runs, n = 11); the pod root (347 runs, n = 10) is a **Supplementary S12** cross-check agreeing in sign and magnitude, not in value.

### 5.3 The dota15 cell: now reproducible, and one wording defect exposed

This cell was not reproducible from the release as first written, and the repair exposed a wording defect (both disclosed, §10 row 2). The repair, the value-by-value agreement and the defect are recorded in **Supplementary S12**.

### 5.4 Two external critiques, checked (one refuted, one judged non-identifiable)

**Supplementary S12** carries the two external critiques in full. We used to write that the gap was simply unidentifiable; the external check **corrects this to** the paired-difference form, which cancels the difficulty term under the pairing assumption stated in §S11. The argument and the recomputation are in **Supplementary S12**.

### 5.5 The same component in third-party public logs

The same quantity appears in other people's logs — a convenience sample with no sampling frame, not a field-level rate: 20 logs from 6 sources found by repository search, median **+1.04 pp**, max **+5.65 pp**, 18/20 positive. The sign is not the informative part: the premium is a within-run maximum minus that run's own final epoch, so it is **non-negative by construction** (both non-positive logs sit at exactly **+0.00**), and the sample's content is its **magnitude**. The queries, the rest of the frame and the reason public logs **cannot establish aliasing** are in **Supplementary S12**.

**Five limitations:** ① the two cells escalated in §5.1 reach  $n = 10$  with **local** runs and test-side evaluations, and the remaining 9 cells' **test** side was closed on 2026-09-30 on one machine; ② **prem\_test** is **best – last**, while the per-epoch test curve now exists for **all thirteen** corpora (**Supplementary S12**, Table S38); ③ **last.pt**'s val recomputation differs from the final **results.csv** row by  $\sim 0.02$  pp; ④ this section does **not** touch §6's 19-benchmark census; ⑤ the aggregate is a **ratio of small means** — **prem\_val** spans **+0.605 to +1.469 pp** across the five arm-cells (Table S20) — so a change of a few hundredths of a point in the numerator moves the rate by tens of points.

### 5.6 Why $\Delta\text{gap}$ 's sign varies by corpus: three of five candidate mechanisms are refuted by the data

**Three of five candidate mechanisms are refuted, and the mechanism itself is not identifiable** — the sign of  $\Delta\text{gap}$  is a sum of same-order terms, so no single mechanism can be read off it (§S11, Remark 3) — **but the paired**

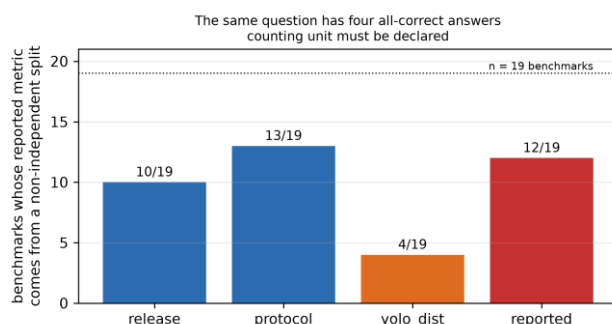
quantity is, and is what we report. The examination, the per-cell components and the exact identity are in **Supplementary S12** and Table S8–S10.

### 6. Finding 3: how general this is across public benchmarks

**Evidence:** `benchmark_split_audit.md`, a line-by-line audit of 19 benchmark rows, plus `split_units_19rows_20260916.md` (recomputation script; no hard-coded percentages).

**The nineteen audited benchmarks**, each read from its own release or dataset paper: COCO [34], PASCAL VOC [21], Objects365 [35], Open Images v7 [36], DOTA v1.0 and v2.0 [37], VisDrone-DET [38], AI-TOD [39], UAVDT [40], xView [41], DIOR [42], NWPU VHR-10 [43], SHWD [44], SFCHD [45], MAFA [46], Mendeley face-mask [22], WIDER FACE [47], CrowdHuman [48] and D-Fire [49] (**per-unit markings:** Supplementary S1) — **the configurations this archive could read, not a sample of the field.** The question is **evaluation validity:** dataset-level bias auditing [50] and classifier-accuracy estimation [51] ask whether a reported number measures what it is taken to measure, but neither audits a release's configuration binding.

#### 6.1 The result under four units (single-rater judgements)



**Fig. 1.** The four counting units on the same nineteen benchmarks — the **printed single-rater** counts (10/19 release, 13/19 protocol, 4/19 yolo\_dist, 12/19 reported); the re-verified tallies are Table 3's second column.

**Table 3.** The four counting units, with denominators named (§3). **Co-headline (2026-10-10):** the same four units were re-coded blind by **five current-generation coders** ( $\kappa$  0.824; **Supplementary S12**), and re-verifying every row first-hand changed **30 cells** — so the table prints **both tallies**: the frozen single-rater counts, and beside them the same four units **re-tallied from the marking table after the 2026-10-09 first-party correction record**, under the **same per-column rule** (the value-to-count mapping is in **Supplementary S12**). Source:

split\_units\_19rows\_20260916.md. Rows marked `n\_a` enter the denominator, not the numerator; alternative readings of the same rows are 12/19 for release, 12/15 the 19 less the four undecidable (n\_a is two rows), and 13/19 for reported under the COCO-val reading. The `clean` row is a reading: the audit counted 5/19, a later pass moved two, so this table reads 3/19. The `reported` row's two denominators: 12-14 of the 19, and 12/15 = 80 % over the fifteen neither n\_a nor undecidable (4 rows are evidence-insufficient). Both ranges span two readings, not intervals, on a single rater's judgement. Independent re-coding reproduces it only at slight-to-fair agreement (Supplementary S12). A model-assisted, non-blind second coding is reliability evidence, not a replacement (Supplementary S12).

| Unit                                                     | non-independent:<br>printed<br>(single rater) | non-independent:<br>after first-party re-verification | of decidable rows: printed → re-verified    | undecidable: printed → re-verified |
|----------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------|---------------------------------------------|------------------------------------|
| ---                                                      | ---                                           | ---                                                   | ---                                         | ---                                |
| **release**<br>(the benchmark's own release)             | **10/19<br>53 %**                             | ≈ **14/19**                                           | 10/19 → **14/19**                           | —                                  |
| **protocol**<br>(the official protocol)                  | **13/19<br>68 %**                             | ≈ **14/19**                                           | 13/19 → **14/19**                           | —                                  |
| **yolo_dist**<br>(the generic YOLO distribution package) | **4/19<br>21 %**                              | ≈ **2/19**                                            | 4/15 ≈ 27 %<br>→ **2/17 ≈ 12 %**            | 4 → **2 rows**                     |
| **reported**<br>(the layer actually reported)            | **12-14 of the 19 ≈ 63-74 %**                 | **10-11 of the 19**                                   | 12/15 = 80 %<br>→ **10/16-11/17 ≈ 63-65 %** | 4 → **2-3 rows**                   |

> **What a count here is, and what a "yes" means.** Every row is a **single rater's judgement**

> (§6.1), so the counts are printed as counts — the percentages beside them are a reading aid, not a

> rate estimated from independent coding. A "yes" under any

> unit means the reported number is **not a sample from an independent split**; it does **not** mean

> the benchmark is broken or the experiment dishonest, and it does not license a generalization claim.

**These 19 rows were coded by one rater** — 28 literal judgements (§10, defect 5). The five blind coders **independently reach 19 of 30 (63.3 %)** corrected cells, eleven unanimously, while **26 of 76** cells stay **undecidable**; a **non-blind, model-assisted** pass agrees at  $\kappa$  **0.874** and **moves two rows** (Supplementary S12). reported means **the number the literature prints**: it is printed as a range, **12–14 of the 19  $\approx$  63–74 %**, with the **80 %** column the lower end over the **12/15** decidable rows, and  $n_a$  rows counted in the **19** but not the numerator (Supplementary S12).

## 6.2 "Same directory" has four levels, and the audit reported only the narrowest

The levels, the rows each admits and the resulting shift are in **Supplementary S12**, Table S12.

**6.3 We do not reproduce the original audit's two numbers, and we say why** one rests on the narrowest reading of "same directory", the other on a stale snapshot; both recomputations are in **Supplementary S12**, Table S13.

## 6.4 One judgment the evidence does not support (corrected)

**One classification correction is recorded in Supplementary S12** (D-Fire moves to the clean group).

## 6.5 Across frameworks: is the convention specific to YOLO?

**Among the pinned default configurations we checked, only the YOLO lineage does this by default.** At pinned commits Detectron2 (TEST.EVAL\_PERIOD = 0), MMDetection, DETR and PaddleDetection **report the final epoch instead** (file-by-file evidence, 25 configuration files: S12, Table S14; **a targeted sample at pinned commits, not a census**, so "the convention" means that family's, not the field's). A **cross-lineage realization test** — a Faster R-CNN (ResNet-50 FPN) detector, a family that **reports the final epoch by default** — on VisDrone under the same three-way split, at **six seeds**, is **positive in all six runs, with a 95 % interval excluding zero (68.2 % realized; +0.0809 pp mean prem\_val, 95 % CI [26.9 %, 123.5 %])** (values, interval and per-run table: **Supplementary S34**): the mechanism is **not confined to the YOLO lineage in this instance**. A **second detector family** (RetinaNet, both corpora) keeps the **direction** but not the strength

(Supplementary S34e). A second corpus, same protocol, same direction: 6/6 and 12/12 runs. This is **one detector, one protocol, two corpora, one setting** — **not a generalisation beyond the YOLO lineage** (Supplementary S34c). Its **68.2 %** is a rate over **six runs** on this probe's own split and is **not comparable** with §8.6's realization rate: different protocol, different denominator.

## 7. Finding 4: what "significant" means, and how the endpoint convention changes it

### 7.1 With $n = 3$ a permutation test cannot reject in principle

A power note for the cells that remain at  $n = 3$ . Recovering each Tier-2 cell's sd from its own printed 95 % CI gives **0.20–0.67 pp**; at  $n = 3$  the required effect is **0.65–2.19 pp** at  $\alpha = 0.05$ , two-sided — **at or above every Tier-1 effect here** (+0.27 to +0.53 pp) — and 0.2–0.7 pp at  $n = 10$ . The nine cells at  $n = 3$  are therefore **not testable at this design**, and are reported descriptively only. A **prefix- $n$  table ( $n = 3 \dots 10$ ) and the derivation with the non-central- $t$  factor ( $3 \cdot 26 \cdot \text{sd}$ ) are in S12**; the permutation floor  $2/2^n$  binds at  $n = 3$ , not the  $t$  test.

For a sign-flip permutation test on  $n$  pairs the smallest attainable two-sided  $p$  is  $2/2^n$ : at  $n = 3$  that is **0.25**, so such a cell has **no resolution at this  $n$** ; after §5.1's escalations no headline cell (Tables 1–2) is  $n = 3$  (the derivation and each  $n$ 's floor: Supplementary S12).

### 7.2 The two clean-protocol cells clear significance by $n = 6$ –8

The weaker endpoint reaches  $*t^*$ -significance at  $n = 6$  on `shwd2sf` ( $p = 0.0164$ ) and  $n = 7$  on `smoke2sf` ( $p = 0.0400$ ); the exact permutation — floored at  $2/2^n$  (0.25 at  $n = 3$ ) — follows one step behind on shwd2sf and none on smoke2sf ( $n = 8$  on both `selected` readings; Table S42 gives every prefix  $n$ , both endpoints). The clean-protocol seed extension that reaches  $n = 10$ , and the per-cell numbers, are in Supplementary S12\*\* and Table S15.

> **First-report attribution (a convention we hold to)**. Three things here belong to the companion

> paper and are cited, not claimed: the two  $\Delta$  values (**+1.434 / +0.507 pp**), the three-way remediation

> protocol itself, and the pre-registered multi-target replication (§9). What this paper reports first is

> \*\*the selection premium (§5), the  $2 \times 2$  realization of that premium on a disjoint split (§8.6–8.7), and

> the formalisation of the realization rate (§3.1)\*\*. The full statement — including how the companion is

> named to the editor while it carries no identifier — is in **Supplementary S12**.

### 7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure)

**One run is recorded with two different endpoints in two archives** — which checkpoint is recorded was never specified: **Supplementary S12**, Table S16. Taking the final-epoch entry (67.12) gives  $\Delta = -1.490$ ,  $p = 0.0141$ . The best-epoch entry (70.06) reads  $\Delta = -0.902$ ,  $p = 0.2931$ , **but had been computed with the run's 'last' reading**; corrected 2026-10-02 it becomes  $\Delta = -1.490$ ,  $p = 0.0141$ , as released. So under the 'best' convention of §5 this cell is **significant**, and nothing printed here moves.

## 8. Finding 5: how far the selection premium transfers to an independent split

### 8.1 The protocol is the companion's; the measurement under it is this paper's

The three-way remediation protocol is **defined and reported in the companion paper** — which first reports that **the effect reproduces in direction while its magnitude falls** — so this section **cites it** rather than presenting the replication. **In this paper's own terms it is three things**: a `val` carved from the pool so that training never sees it; the target corpus's own held-out `test`, with a filename-level check that its images do not intersect training; and an endpoint **defined in writing before any run**. Its **FROZEN-HASH** stamp is stated in §9; the registration `text` itself is the companion's and is **author-side**, so this paper carries the stamp and the readings (**Tables S24–S27**), not the document. What this paper measures

**under** that protocol is the quantity §3.1 formalises: **how much of the 'val'-selected premium survives on a split no selection rule can touch**, and that measurement is this paper's.

### 8.2 The protocol's results on the companion's corpora: transferred

The protocol's results on the companion's corpora are the companion's first report. What this paper adds under the protocol is the measurement of §8.6–§8.7; the transfer (2026-09-21) is recorded in the module-transfer note of that date.

### 8.3 A direct measurement of data-order replication (not seed variance, and not the selection claim itself)

**A direct measurement of data-order replication — reproduced inside this setting:** adding replicates that permute the training file order lowers the mean by 9 %–29 %, each figure taken **against that cell's own  $n = 6$  mean** and not against the pooled headline effect, the **seed-budget discipline** measured rather than assumed ([52]'s setting, §8.4). **What these replicates are.** The per-run randomness is a **deterministic permutation of the training file order** (`--shuffle-seed`, the number in the run's name), **not** the framework's **seed** argument (Supplementary S12, section 9.3, with the two corpora whose replays are identical across `--seed` values). The quantity is therefore **data-order replication**, not seed variance: the 9 %–29 % bounds the **file-order** component of the reported mean, **not** the variance between independent training seeds. That best-of-K selection inflates a reported score is **not new** (§8.6 positions the closest concurrent work, a dose–response over 2,047 datasets), so **what is measured here is the magnitude under a protocol that selects cells for significance first and expands seeds afterwards** (design and recomputation check: Supplementary S12, Table S18). **The title is deliberately the weaker, sample-scoped claim:** with three initializations the interval is wide, and extending to five does not narrow it (the gain SD reaches the data-order component's order), so the components are **consistent with, not established by**, these initializations.

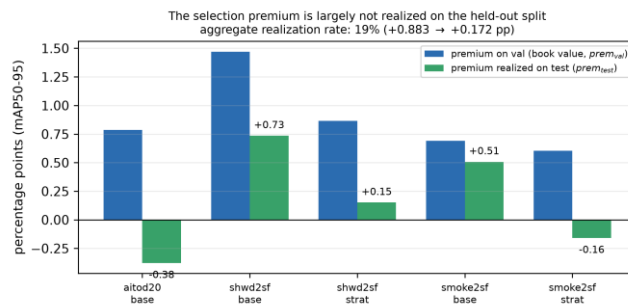
### 8.4 The three variance components: the excluded one is the smaller in this sample — a sample-scoped ordering, not an established one

**The three variance components, and the excluded one is the smaller in this sample:** the single-run data-order component is **0.515 pp**, against initialization and seed components of the same order — the **same ranking** as the shuffle-order result of [52] (Supplementary S12), so we **reproduce the ordering inside this setting** rather than discover it. What was held fixed, and why this is **sample-scoped** — at five initializations the gain SD moves **0.146 → 0.439 pp** and its interval **overlaps** the data-order magnitude, so the two are of the same order rather than one clearly smaller — is in Supplementary S12 and Table S19.

### 8.5 Why "same-pipeline pairing" is a necessity, not a habit

**Pairing is not neutrality:** the paired design covers the **budget and data** axes only, and **not** the optimiser's (why, and the counterexample: **Supplementary S12**).

**8.6 The premium's realization rate: the claim-level interval includes zero** ( $[-17.3\%, 53.2\%]$ ,  $P = 0.118$ ); the point estimates below are descriptive (arm-cell median 17.6 %, five arm-cells,  $n = 10$ ; run-weighted 23.3 %)



**Fig. 2.** The `val`-selected premium against what it realizes on the disjoint `test` split, per arm-cell; the title's aggregate is the plotted pool's — §8.6 names the pool Table 4 uses.

**What this section claims, and what it does not.** The rate below is the realization of the `val`-selected premium on a disjoint split **for the five arm-cells measured here, under the companion's protocol** — **not a constant**. The escalated cells reach a **pooled 55 %**, but the two batches are **not pooled** (different runs, different protocol, §8.1). "The convention realizes about a quarter of the `val` premium" would be false; "in these five cells, about a quarter survives" is what we measured.

**How much of the gain selected on `val` survives on `test`?**

**Table 4.** The premium selected on `val` versus what that same selection action realizes on the disjoint `test` split, per arm-cell (**an unbalanced pool** — two corpora contribute both arms, `aitod20` only its baseline — so the arm-cell is the unit). Source: the **per-epoch `val` curves** (`results.csv`, for `prem_val`), `xeval_20260916/matrix.csv`, and this revision's two batches (`x4_teval.csv` on A, `x4fill_20260925.csv` on B, for `prem_test`). **The §10 memory-pressure bound (defect 10) does not arise here:** its hits are in §3.1's clean batch and §5.1's `p_vistod15`.

|      |     |   |            |                |              |
|------|-----|---|------------|----------------|--------------|
| Cell | Arm | n | `prem_val` | **`prem_test`* | **realizatio |
|------|-----|---|------------|----------------|--------------|

|          |             |       |           | *          | n rate**  |
|----------|-------------|-------|-----------|------------|-----------|
| ---      | ---         | ---   | ---       | ---        | ---       |
| shwd2sf  | baseline    | 10    | +1.469    | **+0.736** | **50 %**  |
| shwd2sf  | strategy    | 10    | +0.865    | **+0.152** | **18 %**  |
| smoke2sf | baseline    | 10    | +0.691    | **+0.509** | **74 %**  |
| smoke2sf | strategy    | 10    | +0.605    | **−0.157** | **−26 %** |
| aitod20  | baseline    | **10* | **+0.766* | **−0.218** | **−28 %** |
|          |             | *     | *         |            |           |
| aitod20  | **strategy* | **10* | **+1.111* | **−0.543** | **−49 %** |
| **†**    | *           | *     | *         |            |           |

> **The inclusion rule, stated as a rule.** An arm-cell enters the pool iff its `test` side reached  $n \geq 5$  (work/item2\_final\_v2\_20260925.py), applied to every cell alike; **exactly five met it at the 2026-09-25 freeze** (50 runs). The aitod20 strategy row **qualifies under the same rule but completed after the freeze**, so it is printed here as the sixth row and kept out of the primary pool — **the `†` row, and the only one; both readings are reported and both intervals cover zero** (Table S20's note, §8.6b).

where  $\text{prem\_val} = V_{\text{max}} - V_{\text{final}}$  (selecting on val) and  $\text{prem\_test} = T_{\text{best}} - T_{\text{final}}$  (realized on test).

**The three bases:** run-level [8.4 %, 37.4 %],  $P = 0.002$ ; five-cell [−17.3 %, 53.2 %],  $P = 0.118$  (claim-level, **zero inside**); eleven-cluster [19.3 %, 60.4 %],  $P < 0.001$  (a bootstrap tail proportion **at the 20,000-resample resolution**, i.e. below the floor this procedure can print).

**Aggregate** (runs weighted equally,  $n = 50$ ):  $\text{prem\_val} +0.879$  pp  $\rightarrow$   $\text{prem\_test} +0.205$  pp. **At the claim level this is not significant:** the **claim-level interval** is the cell-level, five-cluster one, [−17.3 %, 53.2 %],  $P(\text{rate} \leq 0) = 0.118$ , zero inside. Descriptively, arm-cell median **17.6 %** and run-weighted **23.3 %** (span 2/5 negative: Table 4). On a difference scale the eleven clusters give **−1.367 pp**, 95 % CI [−1.59, −1.15] (five-cell denominators **+0.605 to +1.469 pp**). Re-aggregated: run-weighted **36.9 %**, arm-cell-equal **39.3 %**, 95 % CI [19.3 %, 60.4 %] — **completing the cell, not the weighting, is what moves it**: at  $n = 10$  the three weightings agree, so **the fragile axis is pool composition**. **Leave-one-out 9.77 % to 34.16 %**. In two arm-cells the selection action **costs** rather than fails to pay off (smoke2sf strategy, aitod20 baseline); the escalated cells of §5.1 are **not pooled with these five** (different runs, different

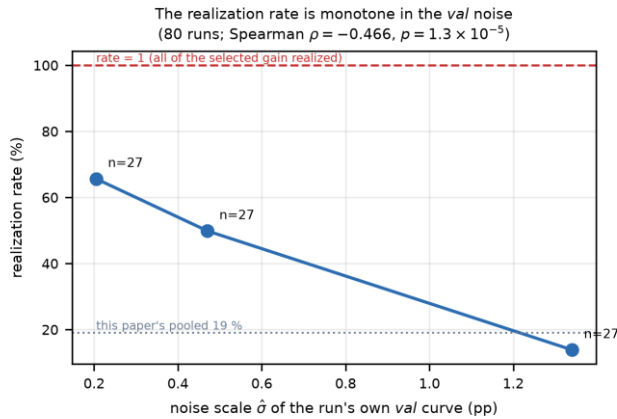
protocol). The closest concurrent work [12], the winner's-curse check, the withdrawn  $\sigma\sqrt{(2 \ln E)}$  form and each cell's corpus pair: **Supplementary S12**.

(the **best.pt**-versus-argmax residual, our withdrawn attribution and why the remainder is **unidentified**: **Supplementary S12**.)

The monotonicity is a correlation, not a causal claim, and at the arm-cell unit the nine-cluster reading is  $\rho = -0.433$  ( $p = 0.244$ ) — see **Fig. 3** and **Supplementary S12**: the  $n = 80$  are the 81 runs with a measured **test** side less the one whose **prem\_val** is 0. It has a named precedent: [53] gives validation overfitting (VO) its name and bounds the expected best observed loss by  $E[\min L] \geq v^* - s_m\sqrt{(2 \log(Nq))}$ , an optimism growing with the number of selection queries  $Nq$ .

**Recomputation status.** The readings were originally **not locally recomputable** (the archive lacks 6 of 7 corpora; §10, **defect 5**); a GPU  $2 \times 2$  **evaluation** on **41** three-way runs of **YOLOv12n** [54]  $\times$  {**best**, **last**}  $\times$  {**val**, **test**} makes them so, **retraining not**, reproducing the archived **best**  $\times$  **test** values to within **0.0047 pp** (median **0.0021 pp**) — the endpoint rule's positive control (**Supplementary S12**).

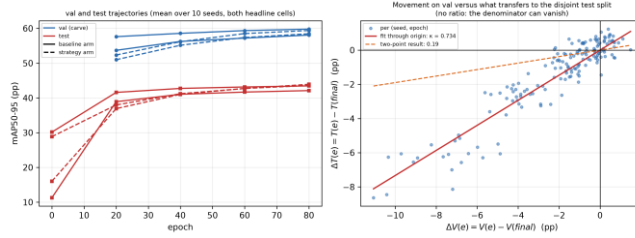
**Three limitations:** ① in the **41-run epoch sweep** only the two headline cells reach  $n = 10$ ; **aitod20** contributes one run of 30 epochs and 6 checkpoints where others contribute **20** (hence 806, not 820), its Table 4 arm-cell a separate  $n = 10$  batch; ② **closed: every retained checkpoint is now evaluated**, the *stability* reading is narrower than the grid; ③  $\kappa$  is relative to each run's own final epoch, not causal. The 806-row evaluation is archived (xeval\_perepoch\_20260918/, with a SHA-256 list).



**Fig. 3.** Realization rate against the noise scale  $\hat{\sigma}$  of the run's own `val` curve, over **80 runs** in three groups of 27 (Spearman  $\rho = -0.466$ ,  $p = 1.3 \times 10^{-5}$ ); dashed: rate = 1, dotted: §8.6's pooled reading.

**Pool composition — and why the claim does not turn on it.** Table 4 prints six measured arm-cells; adding the sixth (`aitod20 strategy`) moves the run-weighted rate **23.3 %**  $\rightarrow$  **8.7 %** and the median **+17.6 %**  $\rightarrow$  **-4.2 %**. **The inclusion rule is a rule, not a per-cell choice:** an arm-cell enters iff its `'test'` side reached  $n \geq 5$ , and exactly five met it at the freeze, where all three weightings coincide (**23.26 %**). The sixth row **qualifies but completed later**, so it is the sensitivity. **Every claim is interval-level and any scope gives the same verdict** — five-cell **[-17.3 %, 53.2 %]** ( $P = 0.118$ ), six-cell **[-4.9 %, +21.3 %]**, eleven-cluster **[19.3 %, 60.4 %]** — the cell-level intervals **both cover zero**, so **no conclusion depends on the pool boundary**.

### 8.7 The per-epoch view: the realization rate is not a constant



**Fig. 4.** **Left:** mean `val` and `test` trajectories over 10 seeds, both headline cells (baseline solid, strategy dashed). **Right:** per-(seed, epoch) `val` movement against what transfers to `test` — the ratio is not taken (its denominator can vanish); the fit through the origin is  $\kappa = 0.734$ .

**One metric self-correction first.** Our first version used  $r(e) = \Delta T(e) / \Delta V(e)$ , producing values such as **-860 %** and **+1031 %**: at late epochs  $V(e) \approx V(\text{final})$ , the **denominator vanishing** — **an ill-conditioned metric, not a finding**. We report only the non-dividing  $\Delta V(e) = V(e) - V(\text{final})$  and  $\Delta T(e) = T(e) - T(\text{final})$ , summarized by the through-origin slope  $\kappa$  of  $\Delta T$  on  $\Delta V$ .

> **Tables S21–S23**  $\rightarrow$  Supplementary §S8.

**Slope  $\kappa$**  (how much of the movement on `val` transfers to `test`). On the **complete curve** —

$n = 765$  (seed, epoch) points over the 41 runs —  $\kappa = 0.708$ ,  $R^2 0.937$ , with a run-clustered (sandwich) 95 % CI [0.671, 0.745]; the within-run estimator gives 0.698. Two controls (Table S43): a per-run  $\kappa$  distribution, and a free-intercept fit whose intercept is indistinct from zero — so the through-origin constraint is supported, not assumed. Across the curve  $\Delta V$  reaches

−8.1 pp and about 71 % transfers: early-training progress is real progress. §8.6's realized share measures the selection-relevant sliver near the peak, where  $\Delta V$  is of order  $\pm 0.2$  pp — so

"most of the movement on `val` transfers" and "the little extra gained by selecting near the peak does not" are both true, and the premium is the residual of the transfer rate. And  $\kappa$  is not constant across arms either — in the direction the selection account predicts: baseline  $\kappa = 0.600$  ([0.581, 0.619]) against strategy  $\kappa = 0.767$  ([0.735, 0.800]), difference +0.168, 95 % CI [+0.130, +0.203].

A second result, and a self-correction the curve forces: the sign-of- $\Delta$ gap claim we wrote from four sampled epochs is too strong for one of the two cells (on the complete curve smoke2sf starts negative) and is narrowed here; per-checkpoint values: Supplementary S11.

The limitations of this section are in Supplementary S12.

## 9. The registered multi-target replication: reported in the companion paper

First report, criterion and verdict stay with the companion paper; nothing here is a claim of this paper. To read its registered table: ① it is a three-way split (train / selection / report); ② it ran under the protocol then in force, where the `val` side aliases the `test` half — the one fact §7 takes; ③ its criterion was frozen before any run existed (FROZEN-HASH 6a7eee7b3e34b15ce5adcba14cf7ea36) and its verdict entered is "criterion not met", deliberately not "hypothesis refuted" — the companion reads the outcome as "direction replicates, size does not". Record: Supplementary S24–S27 (source of truth: the companion's archive).

## 10. Our own defects (disclosed, not hidden)

| #   | Defect                                                                                                                | Handling                                                                                   |
|-----|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| --- | ---                                                                                                                   | ---                                                                                        |
| 1   | The clean-protocol results document <b>**cites</b> a ledger file that does not exist in the sense claimed <b>**</b> : | <b>**We</b> do not point at a replacement file <b>**</b> (renaming would manufacture a new |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                      |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p>`_r10_final_stats.txt` nor the "real log" `_r10_stats_0403.txt` holds <b>**any**</b> headline value (6 probes each, <b>**0/6 hits**</b>) <math>\Rightarrow</math> <b>**no surviving stdout ledger**</b></p>                                                                                                                                                                                                                                                                                                        | <p>false citation); the text says <b>**"no stdout ledger"**, and the p-values are recomputable from</b> <code>`pvalue_table_local.py`</code>. Detail: §S12</p>                                       |
| 2 | <p>~the dota15 cell of <code>`selection_premium.py`</code> was not reproducible~ <math>\rightarrow</math> <b>**fixed**</b> (stale root directory; the matcher assumed a seed suffix the run names do not carry); it now <b>**reproduces M3's +1.12/+0.95/+0.93 value by value**</b></p>                                                                                                                                                                                                                               | <p>Stated in §5.3; the "3 seeds" note was <b>**retracted on 2026-09-21**</b> — the third run's seed is s42, the campaign's fixed shuffle seed. Detail: §S12</p>                                      |
| 3 | <p><b>**`sio_b_results.csv` has no written endpoint convention**</b>: <code>`t2_mask2mende_base100_s42n`</code> is recorded as final (67.12) in one archive and best (70.06) in another</p>                                                                                                                                                                                                                                                                                                                           | <p>Stated in §7.3, with the consequence of each reading (significant versus not)</p>                                                                                                                 |
| 4 | <p>The two headline scripts <b>**originally depended on a rented pod**</b> (<code>`import paramiko`</code> plus a plaintext password) — a paper about evaluation validity whose own analysis is not reproducible from the release</p>                                                                                                                                                                                                                                                                                 | <p><b>**Fixed**</b>: both now <b>**read the local archive**</b>; the main text uses the local versions and lists the pod versions as cross-checks</p>                                                |
| 5 | <p><b>**One script now recomputes the <math>\sigma</math> decomposition**</b> — the data-order and initialization components from the archived per-run values, each asserted against the printed number — but it can only <b>**cite**</b> the augmentation component, whose record holds an SD and no per-run values; <b>**no frozen effect-size rule file**</b>; <b>**no script generates</b> <code>`benchmark_split_audit.md`</code> (hand-written, 28 judgments as literals, though its counts are recomputed)</p> | <p>The first is closed by script, with its one cited (not recomputed) component named; the other two stay labelled "manual / not archived"</p>                                                       |
| 6 | <p>The local archive <b>**has no copy of 6 of the 7 corpora**</b>, and of the three-way carve it holds <b>**only the manifest**</b> (<code>`split_defs_20260916/`</code>), not the other carve directories</p>                                                                                                                                                                                                                                                                                                        | <p><b>**Partially repaired</b> (2026-09-16, GPU run)<b>**</b>: the 41 runs' per-epoch curves, <code>`sio_b_results.csv`</code> and the <b>**<math>2 \times 2</math> evaluation matrix**</b> (164</p> |

|    |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                                                                                                                                                                                                                                                                                                                                    | evaluations) are now local; <b>**retraining remains non-recomputable**</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7  | The audit document has <b>**internal inconsistencies**</b> (nine items, found by our own census)                                                                                                                                                                                                                                                   | Each adjudicated: <b>**3 misreadings by the inventory, 4 real defects, 1 convention risk, 1 classification correction (§6.4)**</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 8  | Our own corpora contain `val == test`                                                                                                                                                                                                                                                                                                              | <b>**That is this paper's argument**</b> , not an embarrassment to be hidden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 9  | <b>**Our first per-epoch report claimed curve-wide stability from a four-point sample**</b> ; the complete curve overturned it for one of the two cells (§8.7)                                                                                                                                                                                     | <b>**Corrected in place, with the retracted sentence quoted**</b> : the claim is <b>**narrowed**</b> , not deleted, and the grid's apparent uniformity is named as partly an artefact of where it sampled. Detail: §S12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 10 | <b>**Memory pressure silently changes the training path**</b> : under GPU memory pressure Ultralytics switches implementation <b>**without changing the recorded configuration**</b> — a batch reduction, or a fallback of label assignment to CPU that can fork the trajectory — and case (a) is <b>**invisible in the standard artifacts**</b> . | <b>**Disclosed, not retracted; we do not call the affected batches “clean”.**</b> Configuration-level hits: <b>**2**</b> places, <b>**4**</b> runs. <b>**A reproducibility residual, not an event size**</b> : same-config repeats are <b>**bit-identical**</b> ( <b>**0.000 pp**</b> , four pairs, same machine and day), while the quantity a bound would be about — archived run against clean re-run — is <b>**0.000 / 0.020 / 0.150 / 0.000 pp**</b> ( <b>**Table S41**</b> ), so <b>**<math>\leq 0.05</math> pp**</b> holds only inside the clean E = 50 batch and is <b>**not extrapolated**</b> ; without a no-event control the event effect is <b>**unidentified**</b> (such a control is itself <b>**not</b> |

reproducible across  
 dates\*\*, \*\*0.440 pp\*\*),  
 and elsewhere it sits  
 inside the zero-event-  
 control spread. The one  
 clear case is  $\Delta V_{\max}$   
 \*\*+0.769 pp\*\*. Event  
 counts, per-run  
 consequences and the  
 log fingerprints are in  
 \*\*Supplementary S12  
 (defect 10)\*\*.

## 11. Recommendations

**Four checks before reporting, each with the artifact that discharges it.** ①

Are `val` and `test` the same source? Make it a **hard pipeline check** on every dataset YAML (§4.2); the output is the resolved paths. ② If you must select on the reported split, say so and report a selection-free endpoint — **the 'final'-epoch endpoint, the only selection-free one this archive measures** — alongside — the step **enlarges** the arm difference (§5.1). ③ State the test convention: below  $n = 6$  a permutation test has no resolution; print the  $2/2^n$  floor beside every p. ④ Declare the counting unit behind every count, as Table 3's header does.

⑤ **A plateau is a post-hoc candidate, not a rule we validate.** Weight averaging is established [55]; what is *measured here* takes the **mean of the ranked top-5 'val' epochs' readings** (a reading average, not a weight average), capturing **70 %** of the realizable gain against the argmax's **30 %**, at the cost of **0.73 pp** below the peak — a post-hoc reading of our own 40 runs, untested as an intervention. The per-rule table and the candidate mechanism are in **Supplementary S11**, Measurement 3.

## 12. Limitations

- The audit reads **split definitions** (YAML, README, paper text), **no datasets downloaded**.
- For several benchmarks the "generic YOLO copy" is **per-project**, not canonical.
- Under the reported unit, **4 rows (AI-TOD, DIOR, Mendeley, D-Fire) are evidence-insufficient**; we do not guess. Counts are **row-equal over the 19**, and the grading enters the **error bar, not the count** (Table 3's second column).
- Two evidentiary differences (pod versus local) are listed side by side.

- The census is a **literature/release survey**, weaker than our **self-measurement** (§3).
- The **difficulty/selection decomposition** (§5.6) covers **2 of the 13 cells** (the headline two); the **identity** behind it is checked on **all 13** (work/gap\_mechanism\_20260916.txt) — what is scoped is the decomposition, not the identity.
- The intervention axis the headline comparisons use is **`lr0`** (**0.001 → 0.005**); two further orthogonal axes were run, one reproducing, as on **lr0**, the **corpus-dependent character** of the shift (**Supplementary S8, Table S40b**); the cross-lineage probe is **one detector, two corpora, and incomparable magnitudes**.
- **Four disclosed uncertainties** bound our claims: the 19-row audit is **single-rater**; the log sample is a **convenience sample**; the GPU memory-pressure effect is **bounded, not estimated** (**≤0.05 pp** where same-config pairs can be compared, fresh replicates being bit-identical; cross-date comparisons carry a **0.440 pp** drift); and the §8.3 replicates vary the **training file order**, not independent seeds. **And, not as an uncertainty: no interval here folds into §8.6's rate. Supplementary S12.**

### 13. Conclusions

A count is meaningless until its **unit** is named: the same nineteen benchmarks answer differently under each declared unit (Fig. 1, Table 3) — **10/19** under release, **13/19** under protocol, **4/19** under yolo\_dist, and **12–14 of the 19** under reported (whose COCO-val reading is **13/19**). **Re-tallied after the 2026-10-09 first-party correction record, under the same per-column rule and with these printed counts frozen, the same four units read 14/19, 14/19, 2/19 and 10–11 of the 19** (Table 3, Supplementary S12) — the two **layer** units rise, the two **package and report** units fall, and the **majority** conclusion is unchanged (the weakest recomputed unit is still **10/19**). Three findings stand. **The most robust number is the transfer slope  $\kappa$  (§8.7): early `val` movement does transfer to `test`, and the premium is its residual.** The premium is positive in all four cells, the baseline gaining more, and the paired  $\Delta$ gap **varies in sign** by corpus — so "the convention understates the effect" is not claimed. On five arm-cells at  $n = 10$  it is realized on a disjoint split at **23.3 %**, interval containing zero. Two things do **not** hold: the  $\sigma\sqrt{(2 \ln E)}$  scaling

was tested and **withdrawn**, and no power analysis was registered, so the nine  $n = 3$  cells stay descriptive. What we ask of a detection paper is narrow: **name the counting unit, freeze in writing which checkpoint is reported, and state the interval at the claim's level**. First report is the companion's (§7.2).

### Declarations

- **Data availability.** The **seven experiment corpora are open**; what the pack lacks is their **images**, so retraining is not recomputable (§10 defect 6). Separately, §5.5's **third-party logs are not redistributed** (hashes + pointers only), as **5 of their 6 sources state no licence**. The package — data, per-epoch matrix, verdicts, checkers, `MANIFEST_sha256.csv` — is at <https://github.com/YBP2005/P2.git> (accessed 2026-09-29). The reproduction package itself — data files, per-epoch matrix, verdict files, checkers, with their SHA-256 list in `MANIFEST_sha256.csv` — is deposited and openly available at <https://github.com/YBP2005/P2.git> (accessed 2026-09-29).

- **Recomputability boundary.** Every reported number is recomputable from the released per-run artifacts **without retraining**, each with its script; **three are not, and they are named**: the corpora and carve directories the archive **holds no copy of** (open corpora, images not in the pack; §10 defect 6), the conflicting-endpoint archives (§7.3), and the `aitod20` completion to  $n = 10$ .

- **Competing interests.** None declared.

- **One quantity, two spellings.** The  $n = 10$  attainable floor is **exactly**  $2/2^{10} = 0.001953$ ; the tables use the exact value, the abstract writes **0.002**. Same floor, spelling only.

- **Reproducibility.** Every number points to a local artifact (**artifact paths in this article and in the Supplementary are relative to the released `02\_release\_data/` directory**); **289 audit anchors** mechanically guard the claims, and six checkers (`audit_anchors_fupaper.py`, `check_crosscite.py`, `verify_pulled.py`, `gap_mechanism_20260916.py`, `verify_pr_forms_20260917.py`, `verify_submission_pack_20260917.py`) ship with the archive: **one runs unaided for a holder of the archive; four are author-side gates; and the remaining one -- the anchor audit itself -- is partly runnable, checking 270 of the 289 anchors from the released copy alone (the other 19 need the authors working-tree inputs)**. The archive is the one deposited above.

## References

> **Note on preprint citations.** Seventeen of the 55 entries give an arXiv edition; **three** also name a peer-reviewed venue and **fourteen** have none. Those fourteen cover four benchmark or dataset canonical releases, two methodological works and eight concurrent or tooling works; **all are cited for positioning only**, and every number comes from our own corpora or the third-party logs of §5.5. The full enumeration is in the supplement.

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